C3.8 Snoop data channel signaling

Table C3-23 shows the signals on the snoop data channel.

Table C3-23 Snoop data channel signals

Signal	Source	Description
CDVALID	Master	Snoop data valid. This signal indicates that the channel is signaling valid snoop data.
CDREADY	Interconnect	Snoop data ready. This signal indicates that the snoop data can be accepted in the current cycle.
CDDATA[cd-1:0][a]	Master	Snoop data. Transfers data from a snooped master.
CDLAST	Master	This signal indicates the last data transfer of a snoop transaction.

a. cd is the width of the snoop data bus.

The standard AXI **VALID**/**READY** handshake rules apply. See *Handshake process* on page A3-37.

If the **CDVALID** signal is asserted, the data value on **CDDATA** and the last transfer indication on **CDLAST**, must not change until **CDREADY** is asserted to indicate the information has been accepted by the interconnect. Once **CDVALID** is asserted, it must remain asserted until **CDREADY** is asserted.

The assertion of **CDREADY** is permitted before, or in the same cycle as, **CDVALID**. If **CDREADY** is asserted before **CDVALID** then **CDREADY** can be deasserted without **CDVALID** being asserted.

The width of the snoop data bus, **CDDATA** is not required to be the same width as the read data and write data buses.

Note

Where the expected cache hit rate is low, and transfer latency is not important, a snoop data bus narrower than the read and write data buses can be implemented.

The snoop data bus can be 32, 64, 128, 256, 512, or 1024 bits wide. However, the following restrictions apply:

- all cache line size transactions must be a full data bus width
- the burst length must be 1, 2, 4, 8, or 16.

These restrictions determine the minimum and maximum data bus widths that can be supported for a given cache line size. See *Cache line size restrictions* on page C3-162.

When **CDVALID** is asserted, all byte lanes of **CDDATA** must be valid, as the snoop data bus does not support byte strobes.

Snoop data is not required for every snoop transaction, it is only provided for a snoop transaction that has a snoop response with the DataTranfer bit asserted. See *Snoop response channel signaling* on page C3-176. When snoop data is required it must be provided in the same order as the associated snoop addresses were presented on the snoop address channel.

All snoop transactions of burst length greater than one are defined to be of burst type WRAP. The order in which data transfers within a snoop burst are provided is the same as for a standard wrapping burst. See *Burst type* on page A3-45.

The **CDLAST** signal must be asserted during the final data transfer associated with a snoop transaction.

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The snoop data channel is optional. However, any cached master that does not support a snoop data channel must still support all snoop transaction types on the snoop address channel.

The cached master must not be required to return dirty data to complete a snoop transaction, and must never use a snoop response with DataTransfer asserted.

To achieve this, the cached master must either:

- not hold dirty data
- must perform a WriteBack or WriteClean, before responding to any snoop process that must obtain a dirty cache line.

Note

This option is not compatible with the WriteUnique and WriteLineUnique transactions. See *Write transactions* on page C4-202.

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